

SYLLABUS :-

1. Name of the subject Hydraulic and Water Resources Engineering Laboratory (New Number) 2. L-T-P 0-0-3 3. Credits 2 4. Name of the Department Department of Civil Engineering 5. Status of the subject (a) Specify the Session: Autumn Semester (b) Please specify the level of the subject: Second year, UG (c) Whether the subject will be offered as core or elective: Core (d) The semester in which the subject will be offered: Autumn (e) Name of the programme in whose curricula this subject will be included: B. Tech. and Dual Degree (CE) 6. Prerequisites for the subject, if any None 7. Objectives To introduce the students to a systematic study of the mechanistic principles of fluid, and their applications to hydraulics through laboratory work. 8. Contents

Experiment Number	Experiment Name
01	Determination of center of pressure on a submerged body
02	Verification of Bernoulli's energy equation
03	Determination of flow properties for forced and free vortices
04	Determination of coefficients of contraction, velocity and discharge for flow through a small orifice
05	Determination of coefficient of discharge of a Venturimeter
06	Study of impact of water jet
07	Determination of efficiency of a Pelton wheel
08	Determination of conjugate depths for a hydraulic jump in a rectangular channel
09	Determination of Darcy friction factor for pipe flow
10	Determination of minor losses in pipes

2. Recommended text books and reference books 1. Fluid Mechanics, SI Version, by Munson, Okiishi, Huebsch and Rothmayer. Wiley 2. Hydraulics Laboratory Manual by Department of Civil Engineering, IIT Kharagpur